

Highlights

Benchmarking LLM Integration Strategies for MCDA-Assisted Household Energy Decision Support

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Highlights

- Bridges natural-language user inputs to rigorous physics-based decision models.
- Hybrid LLM-physics framework achieves τ of 0.88–0.92 and 90–93% Top-1 accuracy.
- Direct and RAG scoring fail near chance ($\tau = 0.01$ –0.31), lacking physics laws.
- Physics calculators narrow model capability gap $4\times$ while cutting costs up to $8\times$.
- Open-source framework scales rapid, expert-grade decarbonization decision support.